

Yeasin Tarek

PhD Candidate — Materials Science & Superconducting Technology | The University of Queensland

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EDUCATION

Doctor of Philosophy

Oct 2022 – Present

The University of Queensland

M.Sc. in Chemistry

Nov 2018 – Jun 2021

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Chemistry

Jun 2014 – Jun 2018

Khulna University

SKILLS

Python

Excel

GSAS

Adobe Suite

MS Office

Origin

LaTeX

Mendeley

HighScore Plus

Dragonfly

Digital Micrograph

PROFILE

PhD candidate in Mechanical and Mining Engineering at The University of Queensland (thesis submitted), specialising in superconducting joint technology for the MRI industry. Doctoral work covers material synthesis, spark plasma sintering, and cryogenic magnetic/transport measurements; process development is under intellectual property protection given its commercial relevance.

Prior research: solar-driven water purification (BCSIR) and biomass-derived Na/K-ion battery materials (BUET). Concurrently Examination Supervisor at UQ. Brisbane-based, open to short-term internships where materials and characterisation experience can support lab operations, hardware validation, and technical reporting.

EMPLOYMENT HISTORY

Research Fellow

Aug 2021 – Aug 2022

Bangladesh Council of Scientific & Industrial Research (BCSIR), Dhaka, Bangladesh

Conducted applied research on solar-powered water purification and desalination systems, including material analysis and process development in an ISO-accredited laboratory.

Postgraduate Research Fellow

Mar 2021 – Jul 2021

Chemistry Department, BUET, Dhaka, Bangladesh

Research on biomass-derived Na/K-ion battery materials — synthesis, electrochemical evaluation, and advanced characterisation.

Experimental Officer

Aug 2020 – Dec 2020

Chemical Engineering Department, BUET, Dhaka, Bangladesh

Chemical and elemental analysis of industrial sludge and effluents using GC-MS, FTIR, and tXRF.

EXPERTISE

- Materials synthesis, processing, and characterisation
- Spark plasma sintering technique
- Cryogenic magnetic and transport measurements of superconducting materials
- Electrochemical analysis: CV, GCD, EIS
- Advanced characterisation: SEM, TEM, XRD, XPS, tXRF, Raman, BET, NMR, AAS, FTIR, GC-MS

Publications

Full record & citation metrics: yatarek.github.io/publications | [Google Scholar](#) | [ORCID](#)

2026

Khan, M. M. A., Saha, S., Mithila, S. N., **Tarek, Y. A.**, Reaz, A. H., & Firoz, S. H. (2026). Reline-assisted synthesis of fcc-hcp Ni/Ni(OH)₂ nanocatalyst for effective reductive hydrogenation of 4-nitrophenol. *Materials Advances*.

Tarek, Y. A., Saha, S., Saha, S., Reaz, A. H., Sharmin, A., Roy, C. K., et al., & Firoz, S. H. (2026). Facile synthesis of self-doped carbon derived from waste tires as an efficient supercapacitor electrode material. *Carbon Letters*, 1–15.

2023

Jamal, A. S. I. M., **Tarek, Y. A.**, Siddique, M. A. B., Shaikh, M. A. A., Debnath, S. C., Uddin, M. R., et al., & Sultana, S. (2023). Development of a fabricated first-flush rainwater harvesting technology to meet the freshwater scarcity in a South Asian megacity, Dhaka, Bangladesh. *Heliyon*, 9(1).

Shakil, R., **Tarek, Y. A.**, Hasan, M. R., Rumon, M. H., Das, R., & Chowdhury, A. N. (2023). Electrospun nanofibers for membrane-based water filtration. In *Electrospun Nanofibrous Technology for Clean Water Production* (pp. 153–179). Springer Nature Singapore. **BOOK CHAPTER**

2022

Tarek, Y. A., Shakil, R., Reaz, A. H., Roy, C. K., Barai, H. R., & Firoz, S. H. (2022). Wrinkled flower-like rGO intercalated with Ni(OH)₂ and MnO₂ as high-performing supercapacitor electrode. *ACS Omega*, 7(23), 20145–20154.

Roy, H., Shakil, R., **Tarek, Y. A.**, & Firoz, S. H. (2022). Study of the removal of basic blue-41 from simulated wastewater by activated carbon prepared from discarded jute fibre. *ECS Transactions*, 107(1), 8407–8420.

Shakil, R., Rumon, M. M. H., **Tarek, Y. A.**, Roy, C. K., Chowdhury, A. N., & Das, R. (2022). Surface-modified nanomaterials-based catalytic materials for water purification, hydrocarbon production, and pollutant degradation.

2021

Barai, H. R., Reaz, A. H., **Tarek, Y. A.**, Saha, S., Mojumder, M. N., Chowdhury, K., et al., & Roy, C. K. (2021). Communication—Self-doped mesoporous activated carbon prepared from car exhaust exhibited long cycle life and high specific capacitance for supercapacitor applications. *ECS Journal of Solid State Science and Technology*, 10(7), 071014.

Roy, H., Rahman, M. M., **Tarek, Y. A.**, & Firoz, S. H. (2021). Encapsulation of industrial cationic pollutants from aqueous solution by nanocrystalline cellulose and its modified forms.

2020

Tarek, Y. A., Reaz, A. H., Shakil, R., Roy, H., Jahan, R. A., Roy, C. K., & Firoz, S. H. (2020). One-step gel formation method for the synthesis of NiO–Mn_xO_y mixed oxide nanomaterials as a prospective supercapacitor material. In *2020 2nd International Conference on Sustainable Technologies for Industry 4.0 (STI)* (pp. 1–6). IEEE. **CONFERENCE PAPER**